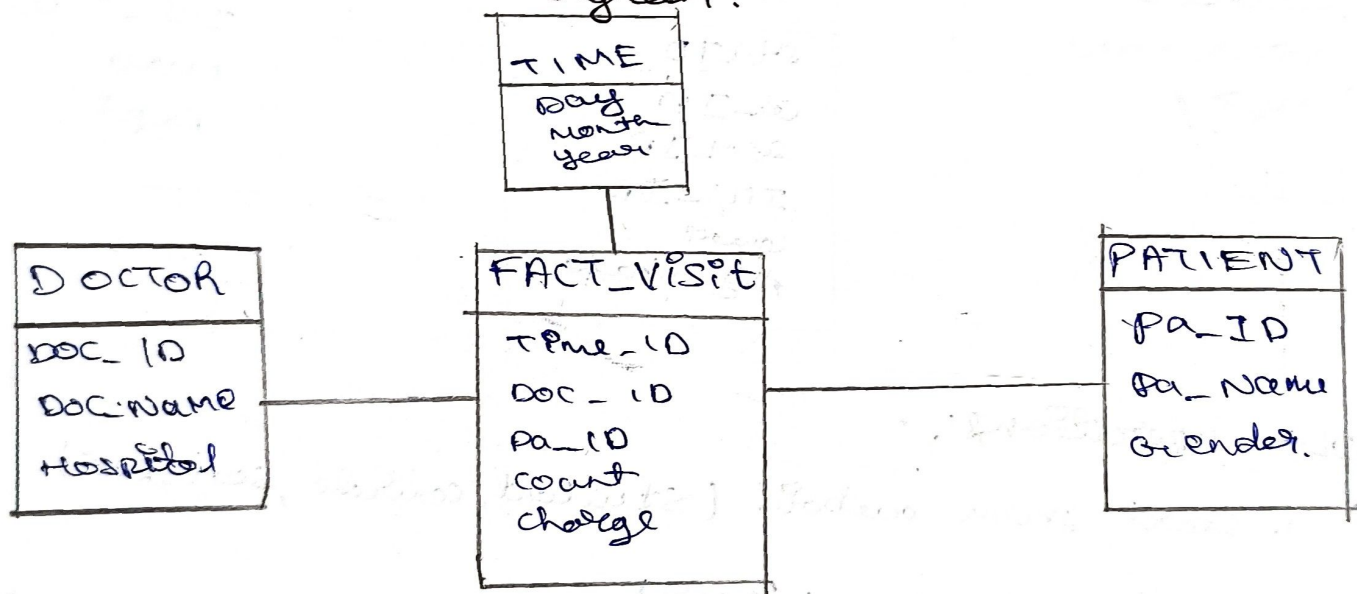


Data warehouse : Doctor - Patient

Three regular schemas used in data warehouse

1. Star schema.
2. Snowflake schema.
3. Fact constellation schema.

Star Schema diagram:



Fact table : FACT_VISIT

Measures:

- count
- charge

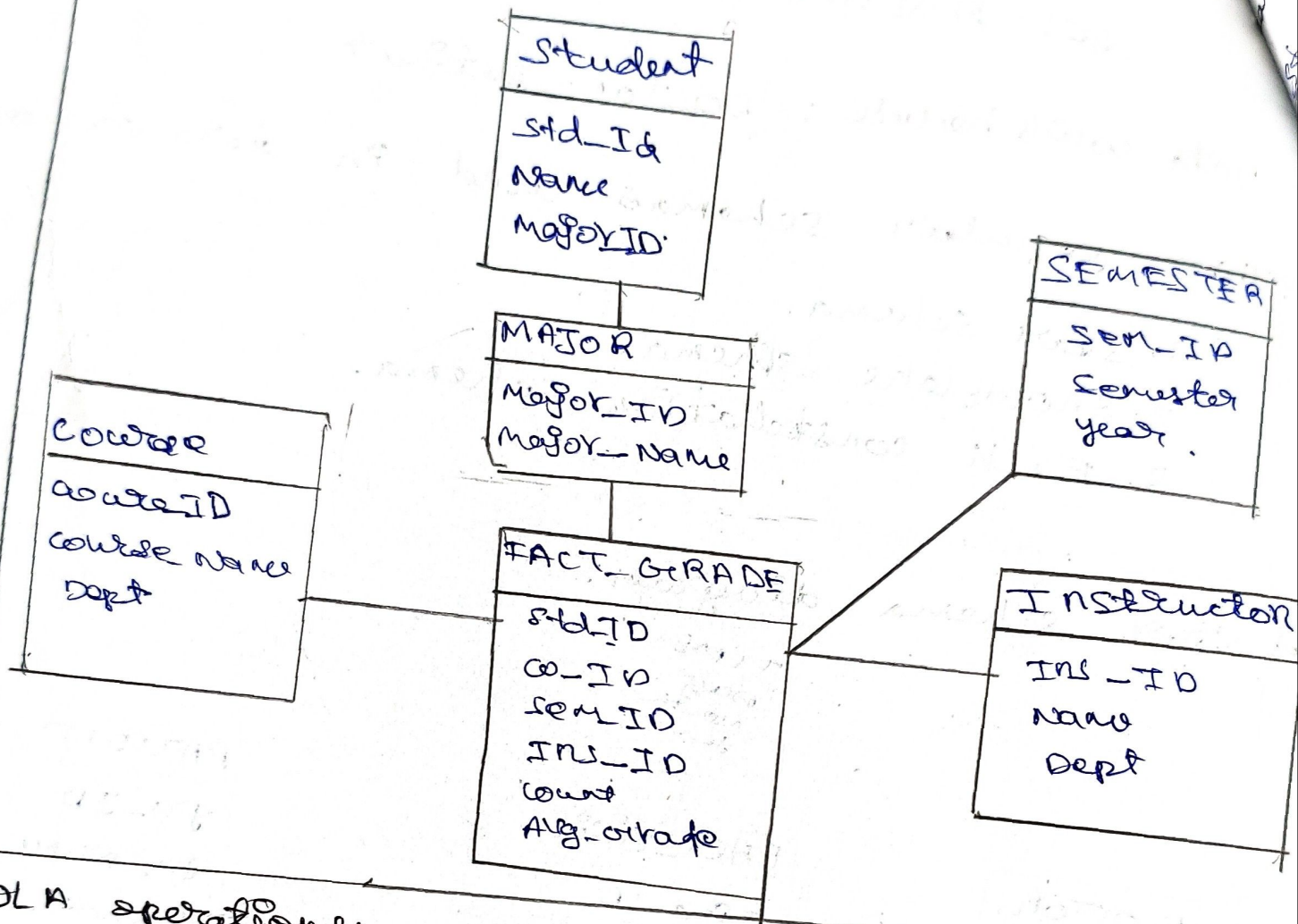
Dimensions:

- Time
- Doctor
- Patient

SQL Query:

select day, month, year, doctor, hospital, patient, count, charge

2. Big university Data warehouse:
of snowflake schema.



OLAP operations:

1. Start from cuboid (Student, course, sem, ins).
2. Roll-up (sem $\rightarrow$ year)
3. slice (course dep = 'CS')
4. Roll-up (Instructor $\rightarrow$ All)
5. Display (Student, CS course Average)

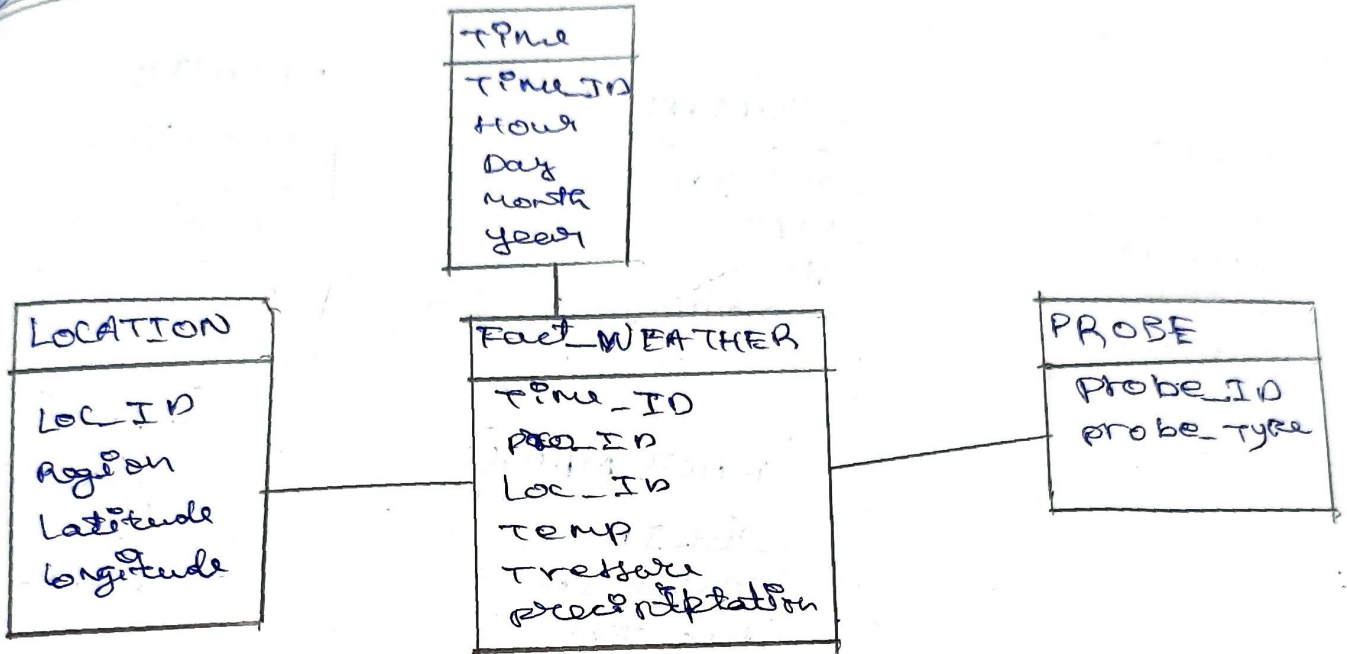
Number of cuboids.

Each dimension has 5 hierarchy levels.
number of dimensions = 4

$$\text{Total cuboids} = 5 \times 5 \times 5 \times 5 = 625$$

625 cuboids.

Weather Bureau Data Warehouse Star Schema



Dimensions

- Time
- probe
- Location

Measures

- Temperature
- Pressure
- Rainfall

This supports:

- Roll-up
- Drill-down
- Slice
- Dice
- weather trend analysis

4. Data cube

Suppose

- n dimensions
- each dimension has p values

a. maximum cells in Base Cuboid
 p^n

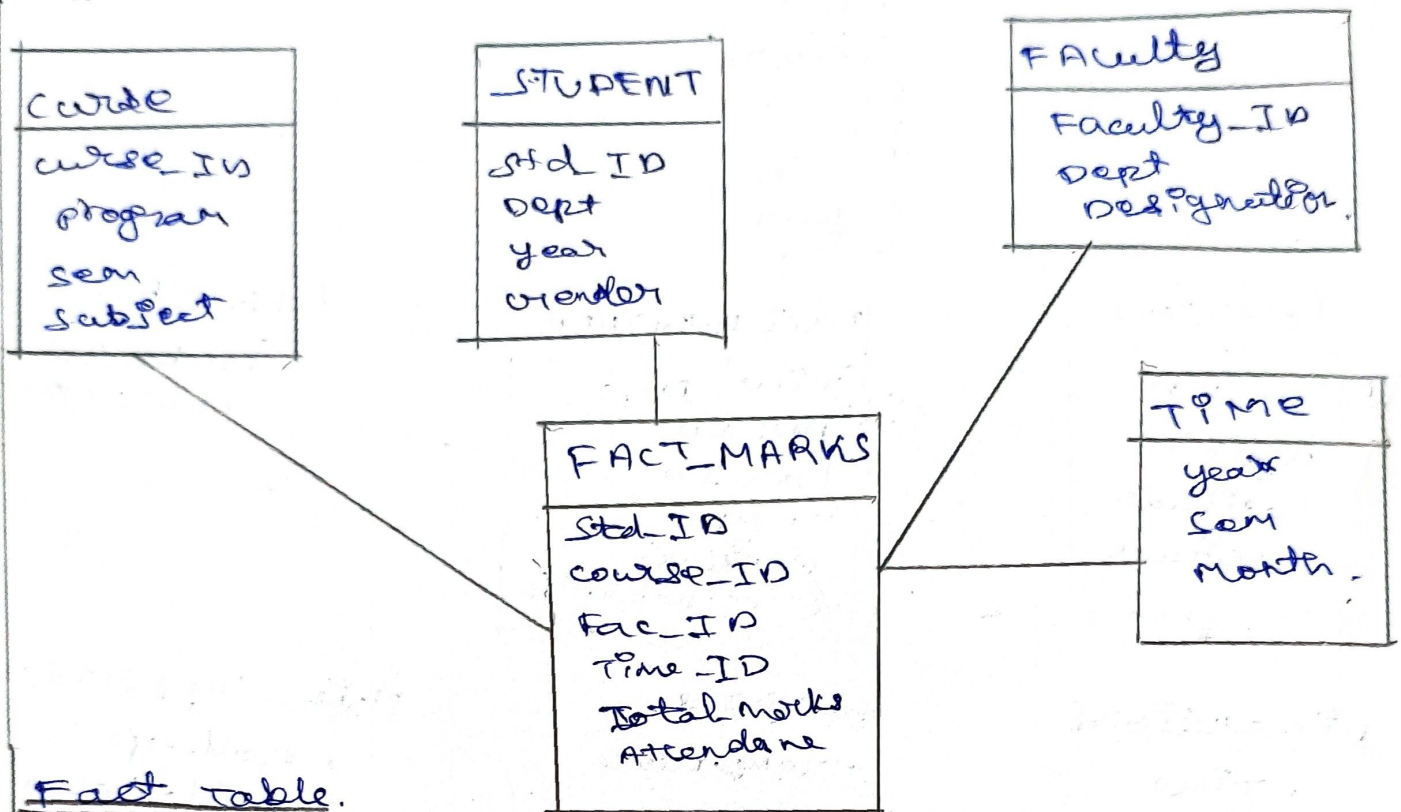
b. minimum cells in Base Cuboid.
1

c. maximum cells in Data cube
 $(p+1)^n$

d. minimum cells in Data cube.
 2^n

5 university Academic performance data warehouse

of star schema:



Fact table.

measures

- Total marks.
- Attendance percentage.

Dimensions

- Student
- course.
- Faculty.
- Time

OLAP operations:-

1. Start from base cuboid. (Student, course, Faculty, Time)
2. Roll-up (Student → department)
3. Roll-up (Time → year)
4. Roll-up (course → All)
5. (Faculty → All)

Number of cuboids:-

$$\text{Student} = 4, \text{course} = 3, \text{Faculty} = 3, \text{Time} = 5.$$

$$= 4 \times 3 \times 3 \times 5.$$

$$= 180$$

180 cuboids